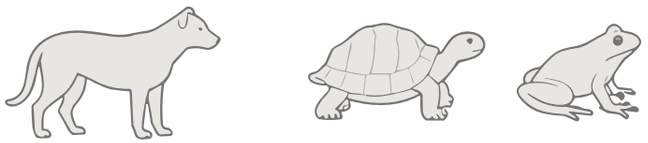
A Design

Placental mammals:
57 species, 1,755 genes

Reptiles & amphibians:
40 species, 1,910 genes



① Reconstruct the ancestral protein

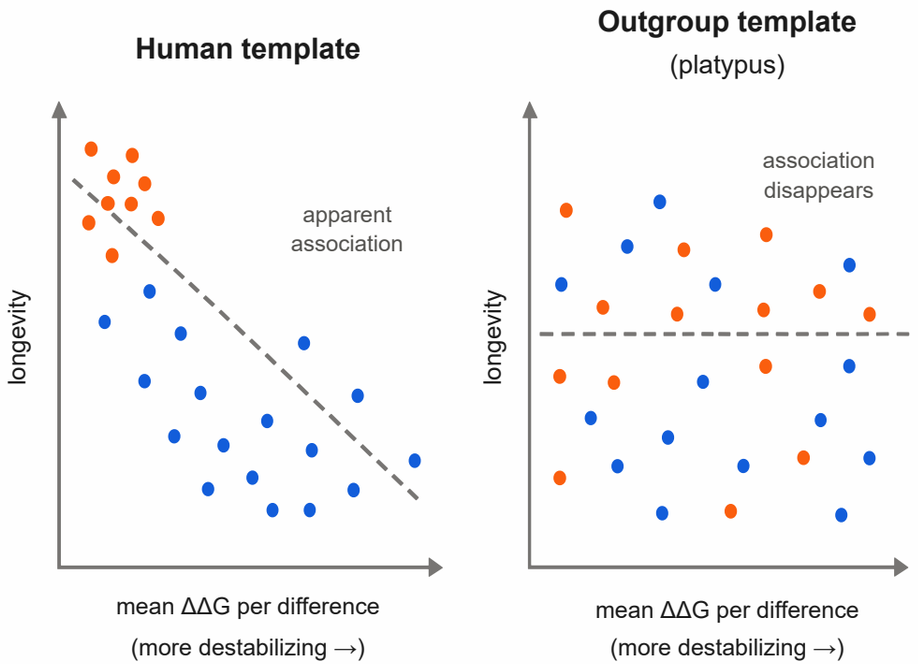
② Find amino-acid differences: ancestor → each species

③ Score each difference on ONE structural template with ThermoMPNN (ΔΔG)

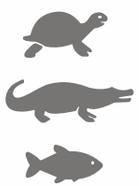
template

④ Mean ΔΔG per difference vs. lifespan (phylogenetic regression)

B The template can create an association

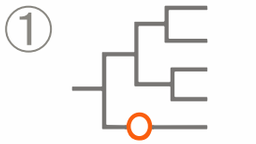


Primate offset: −1.96 SD (human template) → −0.62 SD (platypus template)



Reptiles: a human template favoured slowly evolving turtles and crocodilians; two fish templates (zebrafish, coelacanth) agreed and showed no association.

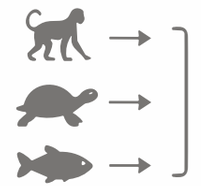
C Four checks



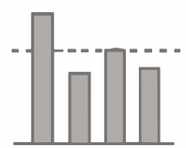
① Measure differences from a reconstructed ancestor, not a modern species



② Use an outgroup template and confirm with a second, unrelated one



③ Test whether signals follow the template's clade or slowly evolving lineages



④ Test dependence on a few clades (bootstrap, clade covariates, within-clade tests)

With these safeguards:
no robust association between predicted protein stability and longevity in tetrapods.